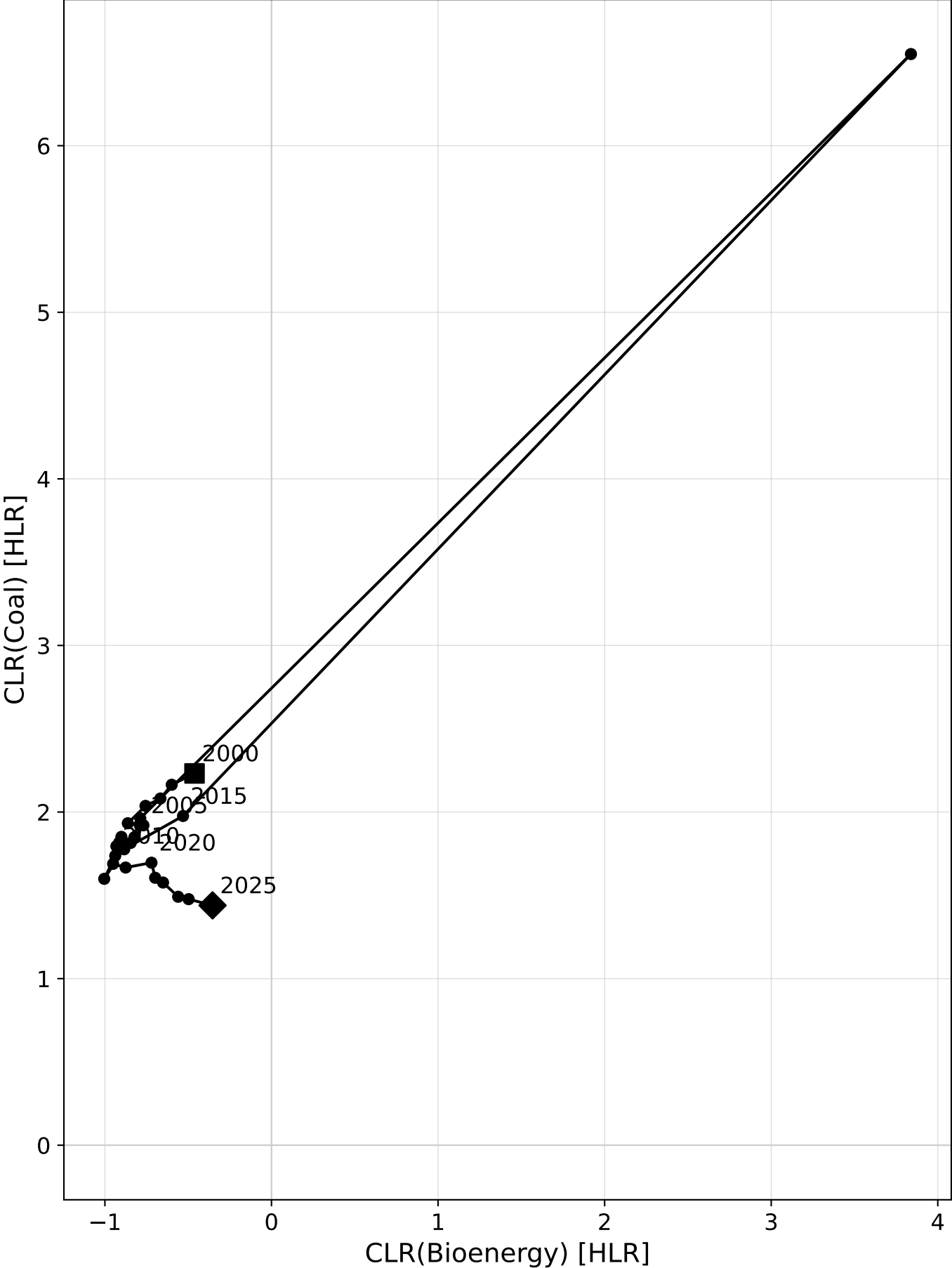
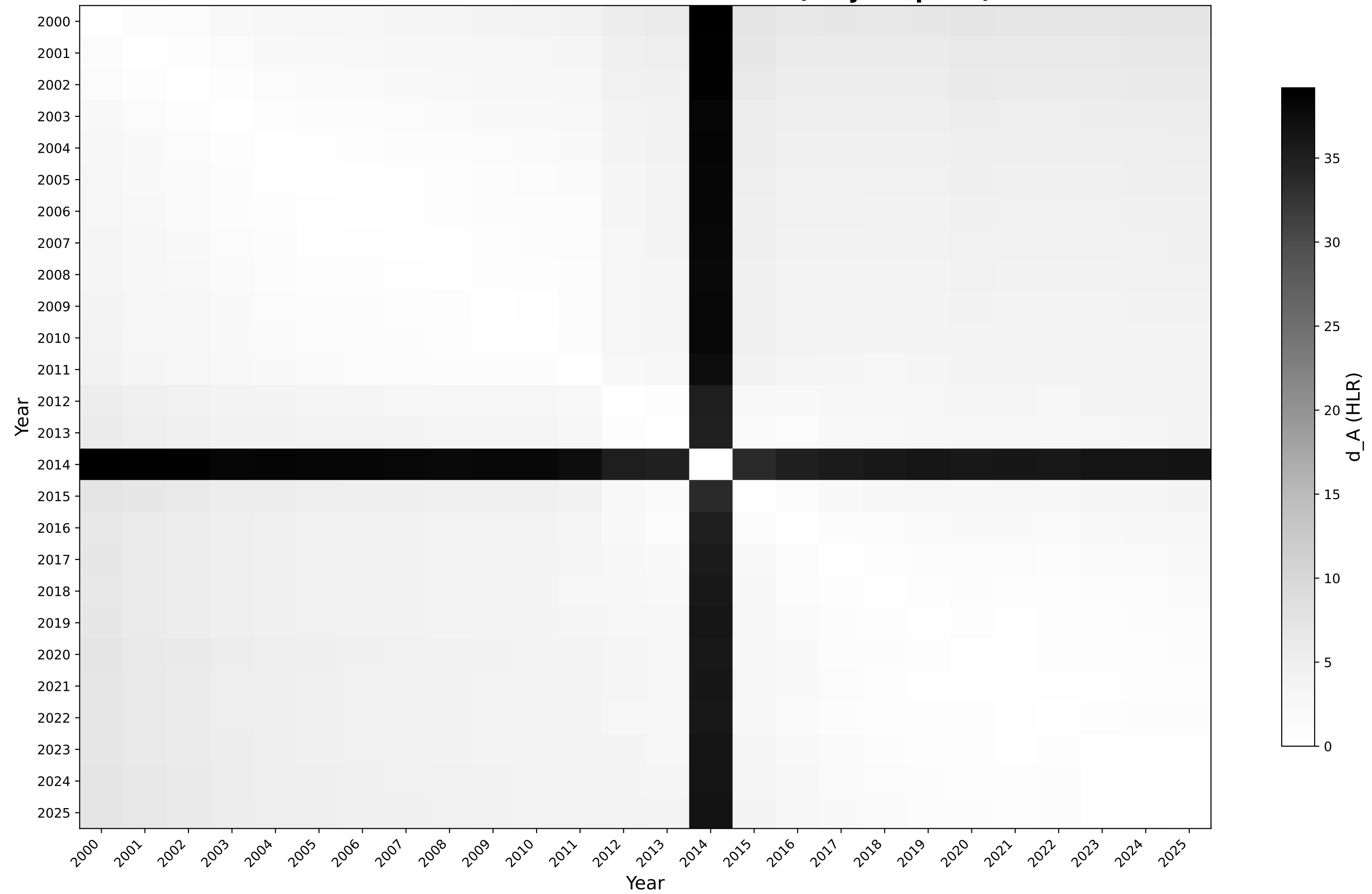


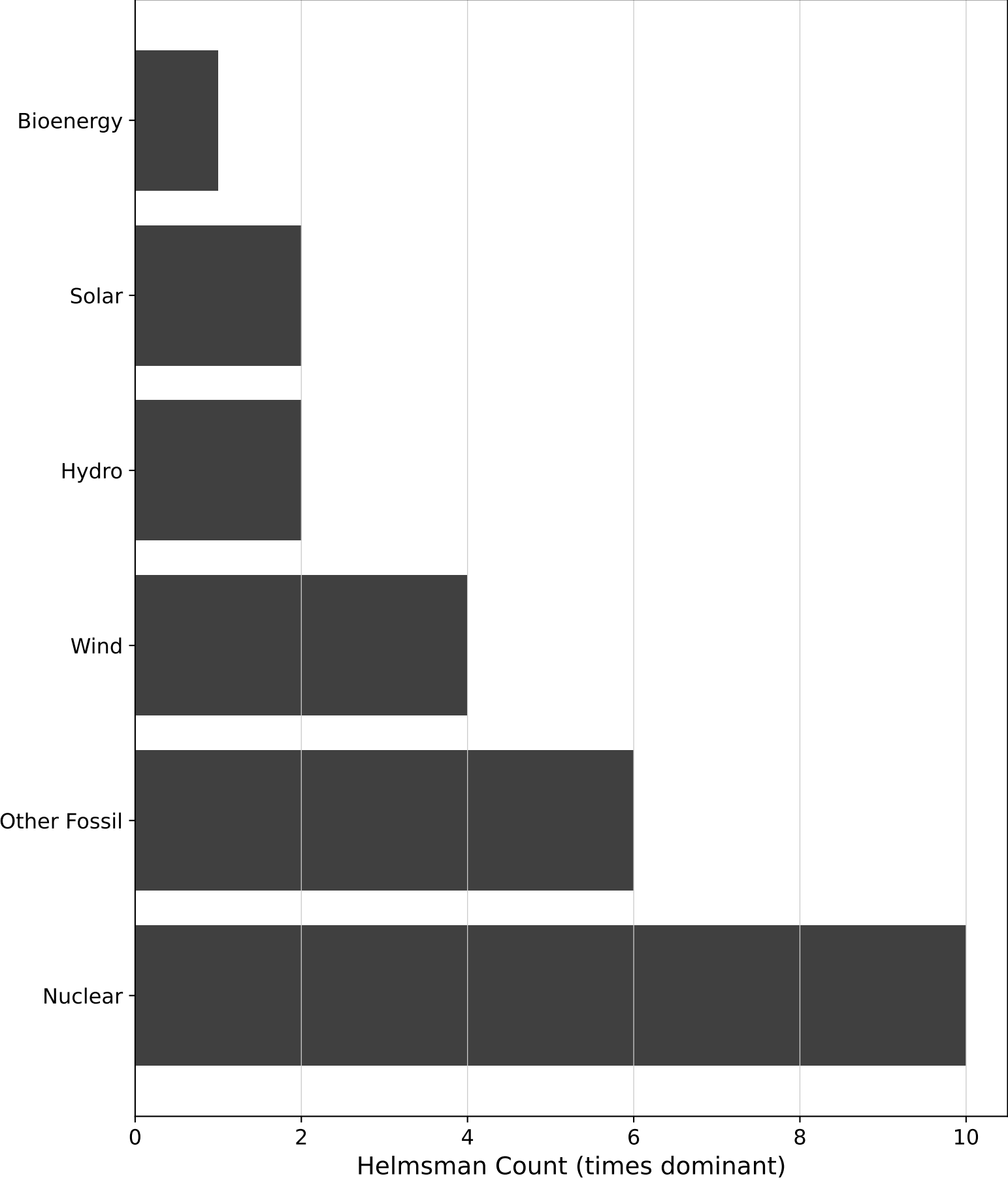
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



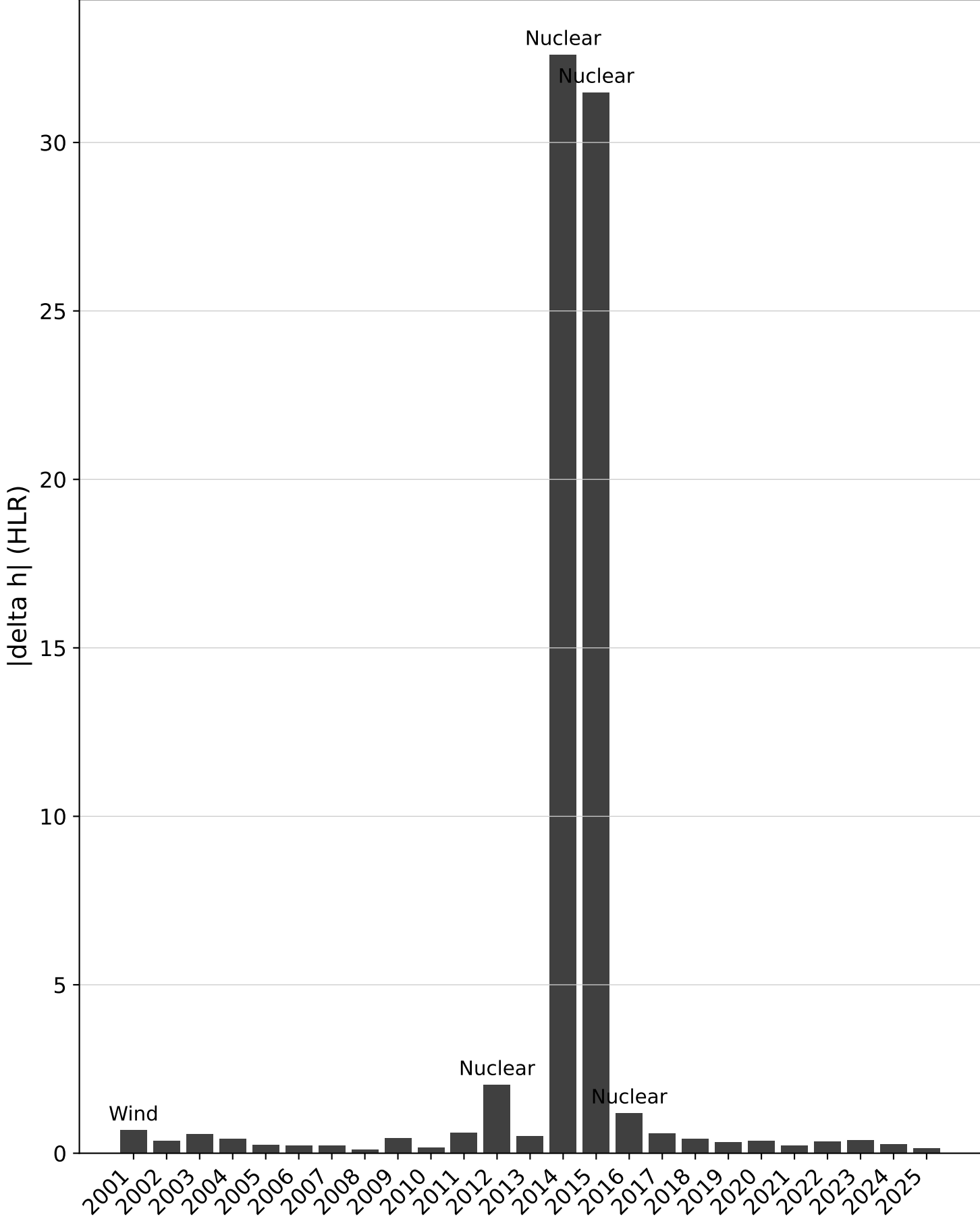
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



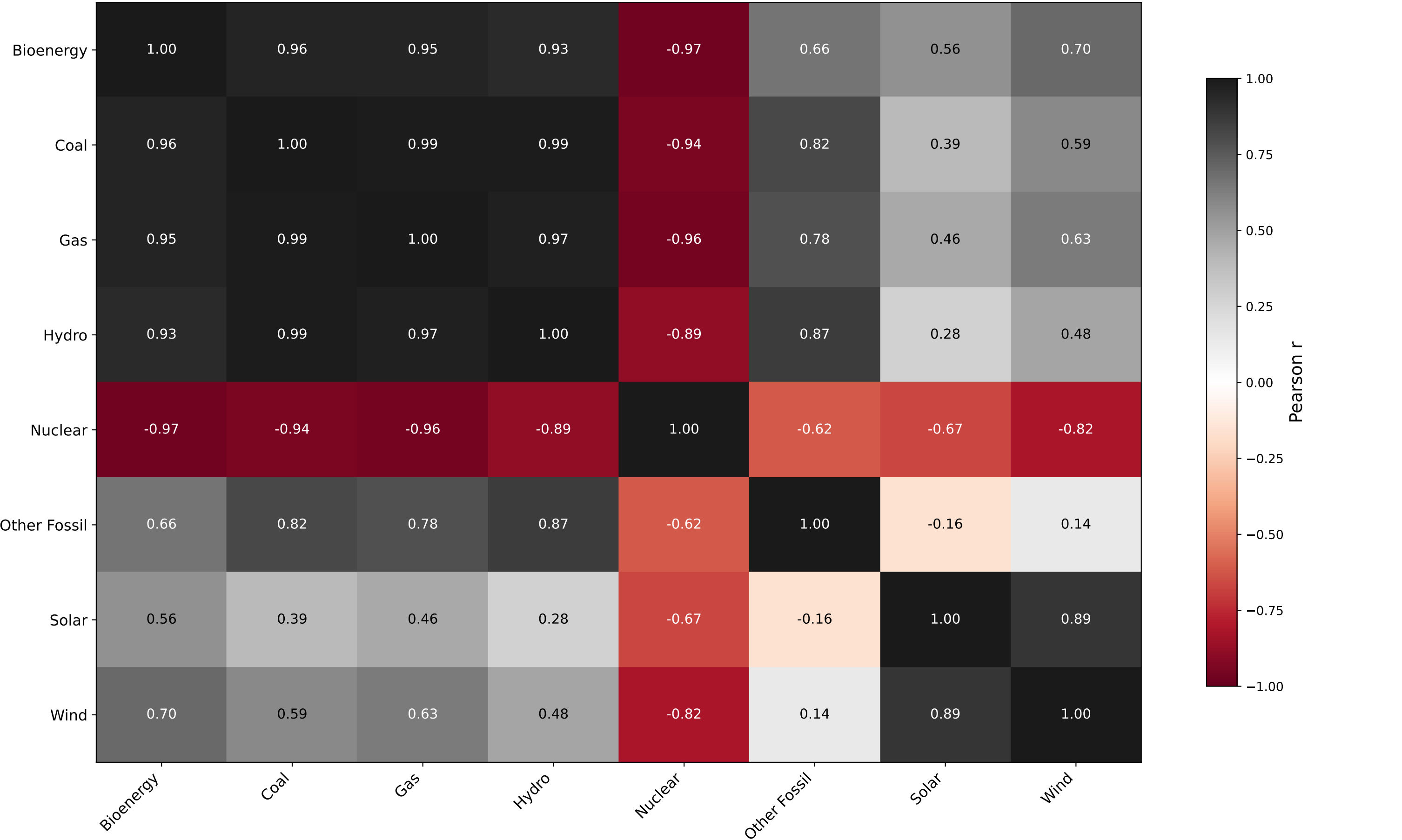
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2000	2014	39.1787	103.1	Nuclear
2	2001	2014	38.9623	103.8	Nuclear
3	2002	2014	38.7724	103.8	Nuclear
4	2004	2014	38.3284	103.6	Nuclear
5	2003	2014	38.2521	101.3	Nuclear
6	2005	2014	38.2438	103.9	Nuclear
7	2006	2014	38.2199	104.3	Nuclear
8	2007	2014	38.0642	103.5	Nuclear
9	2009	2014	38.0444	104.7	Nuclear
10	2010	2014	38.0031	105.1	Nuclear
11	2008	2014	37.9324	102.4	Nuclear
12	2011	2014	37.3369	97.9	Nuclear
13	2014	2025	36.4813	88.9	Nuclear
14	2014	2024	36.4145	87.5	Nuclear
15	2014	2023	36.3177	85.7	Nuclear

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2000	2014	2025	126.1654	5.42
2	2000	2014	2024	124.3973	5.47
3	2000	2014	2023	120.5923	5.58
4	2000	2014	2021	117.8599	5.60
5	2000	2014	2020	117.1553	5.52
6	2000	2014	2022	114.8469	5.62
7	2001	2014	2025	113.0317	5.99
8	2000	2014	2019	111.8742	5.89
9	2001	2014	2024	111.3093	6.04
10	2001	2014	2023	107.6440	6.17

STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.2347	Hs-2	70.6	0.00	0.0000	---	9997
2001	0.2396	Hs-2	58.5	2.82	0.8478	Wind	4333
2002	0.2348	Hs-2	51.8	1.52	0.4967	Wind	2690
2003	0.2228	Hs-2	43.4	3.77	0.7576	Wind	1316
2004	0.2253	Hs-2	37.7	3.33	0.5825	Wind	883
2005	0.2276	Hs-2	34.1	2.48	0.3920	Hydro	707
2006	0.2275	Hs-2	32.2	2.66	0.3162	Other Fossil	582
2007	0.2270	Hs-2	30.0	3.21	0.3717	Hydro	511
2008	0.2323	Hs-2	28.9	2.01	0.2132	Nuclear	457
2009	0.2471	Hs-2	26.2	5.28	0.5460	Other Fossil	365
2010	0.2379	Hs-2	24.4	2.40	0.2823	Solar	294
2011	0.2309	Hs-2	21.5	9.40	0.8378	Nuclear	248
2012	0.3123	Hs-3	21.4	27.46	2.2006	Nuclear	234
2013	0.3126	Hs-3	19.3	7.54	0.6399	Solar	212
2014	0.3409	Hs-3	896.7	66.11	34.8706	Nuclear	1000000000000000
2015	0.3097	Hs-3	20.4	49.67	33.6676	Nuclear	228
2016	0.3030	Hs-3	15.7	16.26	1.3200	Nuclear	199
2017	0.3083	Hs-3	14.0	10.95	0.7679	Other Fossil	173
2018	0.2800	Hs-2	12.4	7.20	0.5079	Nuclear	147
2019	0.2686	Hs-2	12.0	7.62	0.4680	Other Fossil	140
2020	0.2801	Hs-2	12.0	9.79	0.5917	Nuclear	120
2021	0.2468	Hs-2	10.7	4.71	0.3396	Nuclear	108
2022	0.2369	Hs-2	10.2	7.43	0.4247	Other Fossil	110
2023	0.2132	Hs-2	9.7	10.00	0.5551	Nuclear	101
2024	0.2099	Hs-2	9.6	6.04	0.3276	Other Fossil	88
2025	0.1978	Hs-2	9.0	3.76	0.2217	Bioenergy	80